OPERATING SYSTEMS LAB MANUAL

1. Shell Programming: Greatest Among Three Numbers

```
echo "ENTER THREE NUMBERS"
read a b c
if [ $a -gt $b ] && [ $a -gt $c ]
then
    echo "$a is greater"
elif [ $b -gt $c ]
then
    echo "$b is greater"
else
    echo "$c is greater"
fi
```

```
┌──(kali㉿Harish)-[~]
└─$ nano greatest.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x greatest.sh

┌──(kali㉿Harish)-[~]
└─$ bash greatest.sh
ENTER THREE NUMBERS
10 20 30
30 is greater
```

2. Factorial of a Given Number

```
echo "ENTER THE NUMBER:"
read n
fact=1
while [ $n -gt 0 ]
do
    fact=$((fact * n))
    n=$((n - 1))
done
echo "FACTORIAL OF THE GIVEN NUMBER IS $fact"
```

```
┌──(kali㉿Harish)-[~]
└─$ nano factorial.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x factorial.sh

┌──(kali㉿Harish)-[~]
└─$ bash factorial.sh
ENTER THE NUMBER:
```

```
5
FACTORIAL OF THE GIVEN NUMBER IS 120
```

3. Sum of Odd Numbers

```
echo "ENTER THE RANGE:"
read N
i=1
sum=0
while [ $i -le $N ]
do
    sum=$((sum + i))
    i=$((i + 2))
done
echo "SUM = $sum"
```

```
┌──(kali㉿Harish)-[~]
└─$ nano oddsum.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x oddsum.sh

┌──(kali㉿Harish)-[~]
└─$ bash oddsum.sh
ENTER THE RANGE:
2
SUM = 1
```

4. Fibonacci Number

```
echo "ENTER THE LIMIT:"
read n
a=0
b=1
i=1
while [ $i -le $n ]
do
    echo "$a"
    fn=$((a + b))
    a=$b
    b=$fn
    i=$((i + 1))
done
```

```
┌──(kali㉿Harish)-[~]
└─$ nano fibonacci.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x fibonacci.sh

┌──(kali㉿Harish)-[~]
└─$ bash fibonacci.sh
ENTER THE LIMIT:
3
```

```
0
1
1
```

5. Arithmetic Calculator

```
echo "ENTER THE VALUE OF A:"
read a
echo "ENTER THE VALUE OF B:"
read b
echo "ENTER THE OPTION TO PERFORM:"
echo "1. ADDITION"
echo "2. SUBTRACTION"
echo "3. MULTIPLICATION"
echo "4. DIVISION"
read ch

case $ch in
    1) echo "Result = $((a + b))" ;;
    2) echo "Result = $((a - b))" ;;
    3) echo "Result = $((a * b))" ;;
    4) echo "Result = $((a / b))" ;;
    *) echo "Invalid option" ;;
esac
```

```
┌──(kali㉿Harish)-[~]
└─$ nano calculator.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x calculator.sh

┌──(kali㉿Harish)-[~]
└─$ bash calculator.sh
ENTER THE VALUE OF A:
2
ENTER THE VALUE OF B:
3
ENTER THE OPTION TO PERFORM
1. ADDITION
2. SUBTRACTION
3. MULTIPLICATION
4. DIVISION
1
Result = 5
```

6. Largest Digit of a Number

```
echo "ENTER THE NUMBER"
read n
max=0
while [ $n -gt 0 ]
do
    r=$((n % 10))
    if [ $r -gt $max ]
    then
        max=$r
```

```
    fi
    n=$((n / 10))
done
echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

```
┌──(kali㉿Harish)-[~]
└─$ nano largestdigit.sh

┌──(kali㉿Harish)-[~]
└─$ bash largestdigit.sh
ENTER THE NUMBER
7
THE LARGEST DIGIT OF THE NUMBER: 7
```

7. Palindrome or Not

```
echo "ENTER THE STRING TO CHECK PALINDROME"
read str
len=$(echo -n "$str" | wc -c)
i=1
j=$((len / 2))
while [ $i -le $j ]
do
    k=$(echo "$str" | cut -c $i)
    l=$(echo "$str" | cut -c $len)
    if [ "$k" != "$l" ]
    then
        echo "$str is not a palindrome"
        exit
    fi
    i=$((i + 1))
    len=$((len - 1))
done
echo "$str is a palindrome"
```

```
┌──(kali galaxy)-[~]
└─$ nano palindrome.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x palindrome.sh

┌──(kali㉿Harish)-[~]
└─$ bash palindrome.sh
ENTER THE STRING TO CHECK PALINDROME
121
121 is a palindrome
```

8. Reverse of a Given Number

```
echo "ENTER THE NUMBER"
read n
rnum=0
while [ $n -gt 0 ]
do
    remainder=$((n % 10))
    rnum=$((rnum * 10 + remainder))
    n=$((n / 10))
done
echo "REVERSE OF THE NUMBER IS $rnum"
```

```
┌──(kali㉿Harish)-[~]
└─$ nano reverse.sh

┌──(kali㉿Harish)-[~]
└─$ chmod +x reverse.sh
```

```
┌──(kali㉿Harish)-[~]
└─$ bash reverse.sh
ENTER THE NUMBER
122
REVERSE OF THE NUMBER IS 221
```